More Surds

A. $p = \sqrt{6}$ and $q = \sqrt{8}$, show $(pq)^{-1} = \dfrac{\sqrt{3}}{12}$

$$(pq)^{-1} = \frac{1}{\sqrt{6}\times\sqrt{8}} = \frac{1}{\sqrt{48}} = \frac{\sqrt{48}}{\sqrt{48}\times\sqrt{48}} = \frac{\sqrt{16\times3}}{48} = \frac{4\sqrt{3}}{48} = \underline{\frac{\sqrt{3}}{12}}$$

1). $a = \sqrt{2}$ and $b = \sqrt{6}$, show $(ab)^{-1} = \dfrac{\sqrt{3}}{6}$

2). $c = \sqrt{3}$ and $d = \sqrt{6}$, show $(cd)^{-1} = \dfrac{\sqrt{2}}{6}$

3). $e = \sqrt{2}$ and $f = \sqrt{10}$, show $(ef)^{-1} = \dfrac{\sqrt{5}}{10}$

4). $g = \sqrt{3}$ and $h = \sqrt{8}$, show $(gh)^{-1} = \dfrac{\sqrt{6}}{12}$

5). $i = \sqrt{3}$ and $j = \sqrt{15}$, show $(ij)^{-1} = \dfrac{\sqrt{5}}{15}$

6). $k = \sqrt{5}$ and $l = \sqrt{10}$, show $(kl)^{-1} = \dfrac{\sqrt{2}}{10}$

7). $m = \sqrt{5}$ and $n = \sqrt{8}$, show $(mn)^{-1} = \dfrac{\sqrt{10}}{20}$

8). $p = \sqrt{3}$ and $q = \sqrt{18}$, show $(pq)^{-1} = \dfrac{\sqrt{6}}{18}$

9). $r = \sqrt{6}$ and $s = \sqrt{18}$, show $(rs)^{-1} = \dfrac{\sqrt{3}}{18}$

10). $t = \sqrt{6}$ and $u = \sqrt{12}$, show $(tu)^{-1} = \dfrac{\sqrt{2}}{12}$

B. Example: Write $(3 + \sqrt{7})^2$ in the form $a + b\sqrt{7}$, where a and b are integers.

$$\begin{aligned}(3+\sqrt{7})^2 &= (3+\sqrt{7})(3+\sqrt{7})\\ &= 3\times3 + \sqrt{7}\times3 + 3\times\sqrt{7} + \sqrt{7}\times\sqrt{7} = 9 + 3\sqrt{7} + 3\sqrt{7} + 7\\ &= 9 + 6\sqrt{7} + 7 = \underline{16 + 6\sqrt{7}} \quad \text{so} \quad \underline{a = 16,\ b = 6}\end{aligned}$$

1). Write $(1 + \sqrt{3})^2$ in the form $a + b\sqrt{3}$, where a and b are integers.
2). Write $(2 + \sqrt{5})^2$ in the form $a + b\sqrt{5}$, where a and b are integers.
3). Write $(4 + \sqrt{2})^2$ in the form $a + b\sqrt{2}$, where a and b are integers.
4). Write $(3 + \sqrt{5})^2$ in the form $a + b\sqrt{5}$, where a and b are integers.
5). Write $(5 + \sqrt{7})^2$ in the form $a + b\sqrt{7}$, where a and b are integers.
6). Write $(\sqrt{2} + \sqrt{3})^2$ in the form $m + n\sqrt{6}$, where m and n are integers.
7). Write $(\sqrt{3} + \sqrt{6})^2$ in the form $m + n\sqrt{2}$, where m and n are integers.
8). Write $(\sqrt{6} + \sqrt{8})^2$ in the form $m + n\sqrt{3}$, where m and n are integers.
9). Write $(\sqrt{4} + \sqrt{10})^2$ in the form $m + n\sqrt{10}$, where m and n are integers.
10). Write $(\sqrt{6} + \sqrt{12})^2$ in the form $m + n\sqrt{2}$, where m and n are integers.

C. Example: Write $\dfrac{14}{\sqrt{2}} + \dfrac{16}{\sqrt{8}} - 3\sqrt{2}$ in the form $a\sqrt{2}$, where a is an integer.

$$\begin{aligned}&= \frac{14\times\sqrt{2}}{\sqrt{2}\times\sqrt{2}} + \frac{16\times\sqrt{8}}{\sqrt{8}\times\sqrt{8}} - 3\sqrt{2} = \frac{14\times\sqrt{2}}{2} + \frac{16\times\sqrt{8}}{8} - 3\sqrt{2}\\ &= 7\sqrt{2} + 2\sqrt{8} - 3\sqrt{2} = \underline{6\sqrt{2}}\end{aligned}$$

1). Write $\dfrac{40}{\sqrt{8}} + \dfrac{24}{\sqrt{18}} + 5\sqrt{2}$ in the form $a\sqrt{2}$, where a is an integer.

2). Write $\dfrac{18}{\sqrt{12}} + 2\sqrt{3} - \dfrac{15}{\sqrt{75}}$ in the form $b\sqrt{3}$, where b is an integer.

3). Write $5\sqrt{2} - \dfrac{12}{\sqrt{18}} + \dfrac{20}{\sqrt{8}}$ in the form $c\sqrt{2}$, where c is an integer.

4). Write $\dfrac{24}{\sqrt{12}} + \dfrac{15}{\sqrt{3}} - \dfrac{12}{\sqrt{48}}$ in the form $d\sqrt{3}$, where d is an integer.

5). Write $\dfrac{10}{\sqrt{20}} - \dfrac{30}{\sqrt{45}} + 3\sqrt{5}$ in the form $e\sqrt{5}$, where e is an integer.

6). Write $\dfrac{252}{\sqrt{63}} - \dfrac{7}{\sqrt{7}} - \dfrac{14}{\sqrt{28}}$ in the form $g\sqrt{7}$, where g is an integer.

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